Sometimes there are duplicates of a message in the inbox. It could be for a sent message. What possibilities are there for the same message to be displayed, one right after another? In what instances can that happen?

There are several specific reasons and scenarios in the architecture where the exact same message can appear in the inbox, one right after another:

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1. Sending to Self or to Another Site on the Same Server (Sent Copy + Received Copy)

- **How it happens:** In this federated mail system, whenever you send a message, the server saves a record of it into your inbox with the badge **`[ 📤 SENT COPY ]`** (`is_sent_copy = 1`).
- **The duplicate:** If you send a message where one of the destination URLs is:
    1. **Your own website** (e.g. testing the mail form or sending a note to yourself), or
    2. **Another domain on the same server** when using a shared SQLite database: The destination receiver (`inter_site_mail`) receives the HTTP request and inserts the message as an incoming received message (**`[ ✓ RECEIVED ]`**, `is_sent_copy = 0`). A millisecond later, the outbound dispatcher (`send_mail`) finishes and inserts the **`[ 📤 SENT COPY ]`** into the very same table.
- **Result in Inbox:** You see two consecutive items with identical timestamps and message bodies:
    1. Top item: `[ 📤 SENT COPY ]`
    2. Next item: `[ ✓ RECEIVED ]`

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2. Sent Copies Stored with `target_domain = NULL` (Shared Database Leakage)

- **How it happens:** When a message is sent in `index.php`, the query inserting the sent copy into `incoming_messages` does not explicitly set `target_domain`, leaving it as `NULL`.
- **The duplicate:** The inbox query in `get_inbox_messages_for_domain()` searches for:

```
WHERE (target_domain = ? OR target_domain = '' OR target_domain IS NULL)
```

  Because `target_domain IS NULL`, the sent copy matches **every** domain querying that shared database.

- **Scenario:** If Client A and Client B share the database, and Client A sends a message to Client B:
    - Client B receives the incoming message (`target_domain = 'Client B'`).
    - Client A's sent copy (`target_domain = NULL`) also matches Client B's inbox filter.

- In Client B's inbox, both Client A's sent copy and Client B's incoming message appear back-to-back.

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3. Dual Addressing to Client AND Main Contact Point (Subscription Overlap)

- **How it happens:** When a client is subscribed to the Main Contact Point, the client's inbox displays both direct messages and broadcast messages sent to the main contact point (`target_domain IN ($sub_mains)`).
- **The duplicate:** If an incoming message is addressed to **both** the client domain and the main contact point:
  - The client receives a direct copy (`target_domain = 'client.site'`).
  - The main contact point receives a copy (`target_domain = 'main.site'`).
  - If both sites share the same database, the client's inbox query matches *both* records:
    1. Direct message: `[ ✓ RECEIVED ]`
    2. Broadcast copy: `[ ▫ RECEIVED ]` 📢 `Subscribed from main.site`
  - Both records have the exact same body, subject, and sender, appearing consecutively.

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4. Minor URL Variations in the Destination List

- **How it happens:** When composing a mail, the form allows multiple destination URLs. The deduplication logic in `send_mail` checks `in_array($u, $target_urls, true)` with exact string comparison.
- **The duplicate:** If destination URLs differ by a trailing slash (e.g., `https://example.com/contact` and `https://example.com/contact/`):
  - They are treated as two distinct destinations.
  - The sender dispatches two HTTP POST requests to the same endpoint.
  - The remote server receives two dispatches and inserts two identical incoming messages into its inbox back-to-back.

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5. Double-Clicking the Send Button or Browser Form Resubmission

- **How it happens:** If a user double-clicks `[ 🚀 Transmit Federated Mail ]` before the initial request finishes, or if a browser/network retry resubmits the POST request:
  - Two independent requests execute on the server.

- Two outbound transmissions occur and two separate `[ 📤 SENT COPY ]` rows are inserted into `incoming_messages` a fraction of a second apart.

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6. Merging Independent Databases (`merge_databases.php`)

- **How it happens:** When independent SQLite databases from the Main Contact Point and various clients are merged into a single consolidated database:
  - If a message was previously present in both databases (for instance, the main contact point had a copy and a client had a copy, or both had copies of a sent broadcast), deduplication during import checks `(sender_url, received_at, subject, target_domain)`.
  - If one database had `target_domain` set and the other had `NULL` or empty, the merge script treats them as distinct records and imports both into the unified table.

There are also duplications when the main contact point sends mail to subscribers. In what scenarios could that happen?

When the **Main Contact Point** sends mail to subscribers, duplications in the inbox occur primarily due to the intersection of the **single-database shared model**, **automated subscriber forwarding**, and **sent copy storage**.

Here are the specific scenarios in which duplicates happen:

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1. Default Endpoint + Client Endpoint Both in Destination List

- **How it happens:** The send mail form defaults to the Main Contact Point (`https://main.domain/contact`). When the admin at the main contact point wants to send a message to a subscriber, they click `[ + Add Another Destination URL ]` and enter the subscriber's URL (`https://client.domain/contact`).
- **The duplicate:** The message is dispatched to **both** destinations:
    1. **Destination 1 (Client directly):** A message is delivered to `client.domain` with `target_domain = 'client.domain'`.
    2. **Destination 2 (Main Contact Point):** A message is delivered to the main contact point with `target_domain = 'main.domain'`.
- **Result in Client Inbox:** Because the client is subscribed to the main contact point, the client's inbox query fetches direct messages *and* main contact point messages. The client sees **both**:
    1. `[ ✓ RECEIVED ]` (direct message delivered to the client)
    2. `[ ▫ RECEIVED ]` 📢 `Subscribed from main.domain` (the copy sent to the main contact point)

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2. Main Contact Point's Sent Copy Leaking into the Client's Inbox

- **How it happens:** When any user sends mail, a copy is inserted into `incoming_messages` with `is_sent_copy = 1` so it appears in their own inbox. However, `target_domain` is omitted during this insert and defaults to `NULL`.
- **The duplicate:** In a shared database, the client's inbox query looks for:

```
WHERE (target_domain = ? OR target_domain = '' OR target_domain IS NULL)
```

  Because `target_domain` is `NULL` on the main contact point's sent copy, that sent copy matches the client's query too.
- **Result in Client Inbox:** The client sees:

- Copy 1: The incoming message delivered to them (`[ ✓ RECEIVED ]`).
- Copy 2: The Main Contact Point's sent copy (`[ 📤 SENT COPY ]`).

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3. Remote Forwarding Overlap (`is_local = 0` vs Shared DB)

- **How it happens:** Whenever an incoming message is received at the main contact point (`inter_site_mail`), the main contact point automatically forwards the message via HTTP POST to all subscribers in `main_subscriptions` that have `is_local = 0` and a valid `client_url`.
- **The duplicate:**
  - If a client sharing the server's SQLite database was registered or merged into `main_subscriptions` with `is_local = 0` (rather than `is_local = 1`):
    1. The client **already sees** the message directly through the shared database via the virtual broadcast query (`target_domain IN ($sub_mains)`).
    2. Simultaneously, the main contact point executes an HTTP POST to `client_url`. When the client receives this HTTP POST, it inserts **a second physical row** into `incoming_messages` with `target_domain = 'client.domain'`.
  - Both rows appear in the client's inbox one after another.

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4. Direct Sending + Automated Forwarding (External Senders Addressing Both)

- **How it happens:** An external sender or peer site addresses a message to both the main contact point and a subscriber: `Destinations: [ https://main.domain/contact , https://subscriber.domain/contact ]`
- **The duplicate:**
  1. The subscriber receives a direct HTTP POST and saves the message into its inbox.
  2. The main contact point receives its HTTP POST and saves its copy.
  3. The main contact point's automated forwarding loop triggers and forwards the message to all subscribers — sending an HTTP POST back to `https://subscriber.domain/contact`.
- **Result in Client Inbox:** The subscriber receives the message twice over the network: once directly from the original sender, and once forwarded from the main contact point.

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5. Client Subscribed Under Multiple Domain Aliases

- **How it happens:** If a client website is subscribed in `main_subscriptions` under multiple host variants (e.g., both `sub.domain.com` and `www.sub.domain.com`, or with and without port/protocol), or has both a `main_subscriptions` entry and a `site_settings.subscribed_to_main_domain` entry pointing to different variations of the main domain name.
- **The duplicate:** The broadcast query accumulates all variants into the `$sub_mains` list:

```
$where_clauses[] = "(target_domain IN ($sub_placeholders) AND is_sent_copy = 0)";
```

  If messages were stored under different domain variations during installations or migrations, the inbox query can match multiple rows for the same message event.

Summary Table

Scenario	Copy 1 in Client Inbox	Copy 2 in Client Inbox
Both Main & Client URLs in Send Form	Direct message to client (`target_domain = client`)	Broadcast message to main (`target_domain = main`)
Sent Copy `target_domain = NULL` Leakage	Received message (`is_sent_copy = 0`)	Main's sent copy (`is_sent_copy = 1`)
Local Client flagged with `is_local = 0`	Shared DB broadcast query	Forwarded HTTP POST inserted as new row
Sender addresses Main + Client simultaneously	Direct delivery from sender	Forwarded copy from Main

Try to remove duplicate messages. But, for any message sent from the main contact point all subscribers should get a copy in their inbox. However, there shouldn't be a notification for it. If the main contact point as the sender wants the subscribers to get a notification, the main contact point should include itself in the recipient list.

Use To and From for destinations and sources. Don't include https:// or index.php.

All requested changes have been implemented, verified with tests, and committed:

1. Duplicate Message Removal (`db.php`, `index.php`)

- **Database Scoping & Sent Copies:** Sent copies (`is_sent_copy = 1`) in `send_mail` are now explicitly scoped with `target_domain = $current_domain`. In shared database environments, sent copies from one domain can no longer leak into another domain's inbox.
- **Inbox Query Deduplication:** `get_inbox_messages_for_domain()` computes a normalized signature (sender address, normalized subject, body text, and 24-hour time window) to detect duplicates. When multiple entries exist (such as a direct copy and a broadcast copy, or network double-submits), it keeps the most specific record (preserving read status) and discards duplicates.
- **Destination Normalization:** `send_mail` normalizes destination URLs before dispatch, preventing double sends caused by formatting differences (e.g. trailing slashes, `index.php`, or protocol casing).

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2. Main Contact Point Subscriber Delivery & Notification Rules (`index.php`, `db.php`)

- **All Subscribers Receive a Copy:** Whenever the Main Contact Point sends any message, a copy is automatically stored for all local subscribers in the shared database (and dispatched over HTTP to any external remote subscribers).
- **Silent by Default (No Notification):** Subscribed clients do not receive push notifications for messages sent by the main contact point.
- **Self-Inclusion Triggers Notification:** If the Main Contact Point includes its own address (e.g. `tricurse.com/contact`) in the recipient list, this signals intent to notify subscribers. Only in this case are push notifications sent to subscribed clients.

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3. "To" and "From" Clean Formatting (`db.php`, `index.php`)

- **Helper `format_address_display()`:** Strips `https://`, `http://`, `index.php`, and trailing slashes from all displayed endpoints (e.g., rendering `tricurse.com/contact` instead of `https://tricurse.com/contact/index.php`).
- **Form & Modal Labels:**
  - Destination input fields and headers now use **"To:"** instead of "Destination Website URLs".
  - Sender inputs and origin notes now use **"From:"**.
  - Default lists now display **"Default To"** and **"Edit Default 'To' Destinations"**.
- **Inbox & Outbox Displays:**
  - Incoming messages display `From: <Name> (<clean-address>)`.
  - Sent copies and delayed cards display `To: <clean-address>` and `From: <Sender>`.
  - Sent outbox table column is now labeled **"To"** with clean address values.